

Day 19 Report – ARP Poisoning, MITM & Packet Injection

Date: 07 July 2025

Module: Active Sniffing – ARP Spoofing & MITM

Mode: Practical + Theory

Main Focus

Learned and performed **active sniffing techniques** using **ARP poisoning** and **Man-In-The-Middle (MITM)** attacks.

Theory Topics

- **What is ARP Spoofing?**
Manipulating ARP tables to associate attacker's MAC with a legitimate IP.
 - **Man-in-the-Middle Attacks:**
 - Attacker intercepts communication between two parties
 - Can sniff or modify traffic
 - **Packet Injection:**
 - Modifying traffic on the fly (e.g., redirecting pages, injecting payloads)
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Practical Tasks

- Used arpspoof tool in Kali Linux:

```
bash
Copy code
arpspoof -i eth0 -t [victim IP] [gateway IP]
```

- Enabled IP forwarding:

```
bash
Copy code
echo 1 > /proc/sys/net/ipv4/ip_forward
```

- Intercepted traffic with **Ettercap GUI**
 - Observed login credentials on unencrypted HTTP websites
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Key Learnings

- ARP poisoning enables full traffic visibility
 - Tools like Ettercap simplify MITM attacks
 - HTTPS and ARP inspection prevent such attacks
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